

HW#1

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1/27/28

1.10

A) $1305.375_{10} = ?_{16}$ $\underline{Q}$ $\underline{R}$ $.375_{10} = ?_{16}$ $\underline{I}$ $\underline{F}$

$1305 \div 16 =$	81	9	↑	$.375 \cdot 16 =$	6	0 = .6
$91 \div 16 =$	5	1				
$5 \div 16 =$	0	5				

$\Rightarrow 519.6_{16}$

A#2) $519.6_{16} = ?_2$

$5 \quad 1 \quad 9 \quad . \quad 6$
 $0101 \quad 0001 \quad 1001 \quad . \quad 0110 = 010100011001.0110_2$

B) $111.33_{10} = ?_{16}$ $\underline{Q}$ $\underline{R}$ $.33_{10} = ?_{16}$ $\underline{I}$ $\underline{F}$

$111 \div 16 =$	6	15	$.33 \times 16$	5	.28
$6 \div 16 =$	0	6	$.28 \times 16$	4	.48
			$.48 \times 16$	7	.68
			$.68 \times 16$	10	.88
			$.88 \times 16$	14	.08
			$.08 \times 16$	1	.28

$\Rightarrow 6F.547AE_{16}$

B part 2) $6F.547AE_{16} = ?_2$

$6 \quad F \quad . \quad 5 \quad 4 \quad 7 \quad A \quad E \quad .$
 $0110 \quad 1111 \quad . \quad 0101 \quad 0100 \quad 0111 \quad 1010 \quad 1110 \quad 0001 = 01101111.0101010001111010110001_2$

1.11

A) $1011101000.101_2 = ?_8$

$101 \quad 111 \quad 010 \quad 100. \quad 101 = 5724.58 \rightarrow 5 \times 8^3 + 7 \times 8^2 + 2 \times 8^1 + 4 \times 8^0 + 5 \times 8^{-1}$
 $= 2560 + 448 + 16 + 4 + .625$
 $= 3028.625$

$1011 \quad 101 \quad 0100. \quad 1010 = BD4.A_{16} \rightarrow 11 \times 16^2 + 13 \times 16^1 + 4 \times 16^0 + 10 \times 16^{-1}$
 $= 2816 + 208 + 4 + .625$
 $= 3028.625$

$$\begin{array}{r}
 11 \cdot 17 \\
 \text{A): } \begin{array}{r}
 1111 \\
 + 1001 \\
 \hline
 11000_2
 \end{array}
 \end{array}$$

$$\begin{array}{r}
 1111 \\
 - 1001 \\
 \hline
 0110_2
 \end{array}
 + \begin{array}{r}
 1111 \\
 + 0000 \\
 + 01111 \\
 + 0000 \\
 \hline
 1001111
 \end{array}$$

$$\begin{array}{r}
 1111 \\
 100,00111_2
 \end{array}$$

$$\begin{array}{r}
 \text{B): } \begin{array}{r}
 1101001 \\
 + 0110110 \\
 \hline
 10011111_2
 \end{array}
 \end{array}$$

$$\begin{array}{r}
 00100101 \\
 - 0110110 \\
 \hline
 0110011_2
 \end{array}$$

$$\begin{array}{r}
 1101001 \\
 + 0000000 \\
 + 1101001 \\
 + 11010010 \\
 + 11010010 \\
 + 1001110110 \\
 + 0000000 \\
 + 1001110110 \\
 + 101001 \\
 + 100100000110 \\
 + 1101001 \\
 \hline
 1011000100110_2
 \end{array}$$

$$\begin{array}{r}
 \text{A): } \begin{array}{r}
 00101110_2 \leftarrow \text{quotient} \\
 10111101001 \\
 - 101 \\
 \hline
 0000 \\
 - 101 \\
 \hline
 000000 \\
 - 101 \\
 \hline
 000010 \\
 - 101 \\
 \hline
 00000011_2 \leftarrow \text{Remainder}
 \end{array}
 \end{array}$$

$$\begin{array}{r}
 \text{B): } \begin{array}{r}
 000100011_2 \leftarrow \text{quotient} \\
 101110000011 \\
 - 1011 \\
 \hline
 00000001 \\
 - 1011 \\
 \hline
 000001001 \\
 - 1011 \\
 \hline
 00100_2 \text{ Remainder}
 \end{array}
 \end{array}$$

1.31

$$\begin{array}{l} 2 = 0010 \\ 5 = 0110 \end{array} \left. \begin{array}{l} 1100 \\ 0100 \end{array} \right\} \begin{array}{l} 011 \\ 011 \end{array} \Rightarrow \boxed{93_{10}}$$

1.37 (2's complement)

A): 01001
 $-11010 \rightarrow 00101$
 Check: 9
 $-(6) = 15$
 00101
 00110
 $+01001$
 $\boxed{01111}$

B): 11010
 $-11001 \rightarrow 00110$
 00110
 $+11010$
 $\boxed{10001}$ only 5 bits needed

check: 00101
 $00110 \rightarrow -6$
 $00111 \rightarrow -5$

C): 10110
 $-01101 \rightarrow 10010$
 10010
 $+10110$
 $\boxed{10100}$ only 5 bits needed

D): 11011
 $-00111 \rightarrow 11000$
 11000
 $+11011$
 $\boxed{110100}$ only 5 bits needed
 001011
 001100
 $= -12$

check: 01001
 $01010 \rightarrow -10$
 -13
 $= -26$
 10100 only 5 bits needed
 overflow because
 2 negatives added to
 positive

E): 11100
 $-10101 \rightarrow 01010$
 01010
 $+11100$
 $\boxed{100111}$ only 5 bits needed

check: 00011
 $00100 \rightarrow -4$
 01010
 $01011 \rightarrow +11$

$$11010 \rightarrow -5$$

0 1 10 / 3 > (1's complement)

A): $\begin{array}{r} 0110 \\ \times 0110 \\ \hline 0110 \\ 1100 \\ \hline 01111 \end{array}$ 9
 $-11010 \rightarrow -(-6) = 5$
 $\boxed{01111}$

B): $\begin{array}{r} 0110 \\ \times 1100 \\ \hline 0110 \\ 1100 \\ \hline 00001 \end{array}$ $\rightarrow 00101 = -5$
 $-11001 \rightarrow 00110 = -(-6)$
 $\boxed{00001}$

C): $\begin{array}{r} 0100 \\ \times 1010 \\ \hline 0100 \\ 1000 \\ \hline 01101 \end{array}$ $\rightarrow 01001 = -9$
 $-01101 \rightarrow -13$

D): $\begin{array}{r} 0100 \\ \times 1101 \\ \hline 0100 \\ 1100 \\ \hline 00111 \end{array}$ $\rightarrow 00100 = -4$
 $-00111 \rightarrow -7$

$\boxed{01001}$ ← overflow because sign is + instead of -

$\boxed{10100}$ → check: $01011 = 11$

E): $\begin{array}{r} 0101 \\ \times 1100 \\ \hline 0101 \\ 1100 \\ \hline 00101 \end{array}$ $\rightarrow 00011 = -3$
 $-10101 \rightarrow 01010 = -(-10) = 10$

$\boxed{00111}$